

BG

BIGLANG GWAPO BARBERSHOP

Quezon City, Metro Manila

SYSTEM DOCUMENTATION

Barbershop Management & Booking System

Version 1.0 • March 2026

Document Information

Project: BG Biglang Gwapo Barbershop

Type: Web-Based System

Version: 1.0

Date: March 2026

Status: Production Ready

Tech Stack

- Frontend: HTML5, CSS3, Bootstrap 5, JavaScript
- Backend: PHP 8.2
- Database: MySQL / MariaDB
- Email: PHPMailer + Gmail SMTP
- Security: reCAPTCHA v3, CSRF, Rate Limiting
- Server: Apache / XAMPP (localhost)

1. System Overview

The BG Biglang Gwapo Barbershop System is a full-stack web application designed to digitize and streamline the operations of a barbershop business in Quezon City. It covers online appointment booking, customer management, walk-in queue management, staff management, reporting, and administrative controls.

1.1 System Goals

- Allow customers to book appointments online anytime, anywhere
- Provide real-time barber availability and slot management
- Give administrators a complete dashboard to manage all business operations
- Automate email notifications for bookings, reminders, and cancellations
- Maintain secure access with multi-layer security implementation

1.2 System Architecture

Pattern	MVC-inspired PHP with procedural includes
Database	MySQL (biglang_gwapo) — 13+ tables
URL Structure	localhost/bg-barbershop/
Root Path	C:/xampp/htdocs/bg-barbershop/
Session	PHP native sessions with security hardening
Email	PHPMailer 7.0 via Gmail SMTP (App Password)

1.3 Folder Structure

Folder / File	Purpose
views/user/	Customer-facing pages (homepage, booking, gallery, etc.)
views/admin/	Admin panel pages (dashboard, appointments, reports, etc.)
includes/	Shared PHP partials (header, navbar, footer)
includes/admin/	Admin-specific partials (header, footer, queue card, etc.)
api/	Backend API endpoints (booking, get_slots, etc.)
api/admin/	Admin-only API endpoints (update_appointment, export, etc.)

Folder / File	Purpose
config/	Configuration files (db.php, mail.php, mailer.php, security.php)
assets/css/	Stylesheets (style.css, pages.css, admin.css)
assets/js/	JavaScript files (main.js)
assets/images/	Local images (barbers, gallery, logo)
cron/	Scheduled tasks (send_reminders.php, run_reminders.bat)
vendor/	Composer packages (PHPMailer)

2. Database

Database name: biglang_gwapo. The system uses MySQL/MariaDB with InnoDB engine. All tables use UTF-8 character encoding.

2.1 Database Tables

Table	Description	Key Columns
admins	Admin panel users	id, name, email, password, role, is_active
customers	Registered customers	id, first_name, last_name, email, phone, password, is_active
barbers	Barber profiles	id, first_name, last_name, role_title, bio, photo, instagram, facebook, is_active, sort_order
service_categories	Service groupings	id, name, description, sort_order
services	Individual services	id, category_id, name, description, price, duration_mins, is_featured, is_active
appointments	All bookings	id, reference_no, customer_id, guest_name, guest_phone, barber_id, appointment_date, start_time, end_time, status, reminder_sent
appointment_services	Services per booking	id, appointment_id, service_id, price_snapshot
blocked_slots	Blocked dates/times	id, barber_id, block_date, start_time, end_time, reason
walk_in_queue	Walk-in queue	id, customer_name, barber_id, service_id, status, started_at, completed_at
gallery	Gallery photos	id, title, image_path, category, is_featured, is_active
testimonials	Customer reviews	id, customer_name, location, review_text, rating, is_featured, is_active
settings	System settings	id, setting_key, setting_val, label
schedules	Barber schedules	id, barber_id, day_of_week, start_time, end_time, is_available

2.2 Default Admin Credentials

Change these credentials immediately after first deployment.

Email	admin@bgbarbershop.com
Password	Admin@123 (change after first login)
Role	superadmin

3. User-Facing Pages

All user pages are located in `views/user/`. They are dynamically loaded from the database and are mobile-responsive.

3.1 Homepage (`index.php`)

The main landing page loaded through the root `index.php` file.

Section	Description
Hero	Full-screen banner with CTA — Book Appointment and Our Services buttons
About	Brief shop introduction with stats (years, barbers, clients, rating)
Services	Dynamic — 6 service categories loaded from database with starting prices
Our Barbers	Dynamic — Active barbers from database with photos and social links
Gallery	Dynamic — Featured photos from gallery database table
Testimonials	Dynamic — Featured reviews from testimonials table
Contact	Contact info, Google Maps embed, and quick contact form

3.2 Booking System (`booking.php`)

4-step multi-page booking form with real-time validation and conflict detection.

Step	Page	Description
Step 1	Services	Checkbox selection with conflict detection. Max 1 per category. Packages conflict with all other services.
Step 2	Barber	Dynamic barber cards from DB. Shows availability status. Option for No Preference.
Step 3	Schedule	Date picker + time slots (9AM–7PM, 30-min intervals). Booked and blocked slots are grayed out automatically.
Step 4	Your Info	If logged in: pre-filled name/email, phone input only. If

Step	Page	Description
		guest: login prompt + full form. Submit sends to API.

After successful booking, the system redirects to booking_confirmation.php with the reference number.

3.3 Customer Account Pages

Page	File	Description
Login	login.php	Email + password login with brute force protection. Show/hide password toggle.
Register	register.php	New customer registration with full validation.
My Bookings	my_bookings.php	Lists all customer bookings. Can cancel pending/confirmed future appointments.
Profile	profile.php	Edit personal info (name, email, phone) and change password. Booking stats summary.
Booking Confirmation	booking_confirmation.php	Printable booking receipt with all appointment details and shop info.
Logout	logout.php	Clears customer session only (admin session preserved).

3.4 Other Pages

Page	File	Description
Services	services.php	Full service menu loaded dynamically from DB grouped by category with prices and duration.
Gallery	gallery.php	Masonry grid with filter by category. Lightbox on click. Dynamic from gallery DB table.
About	about.php	Shop story, values, stats strip, and team section loaded from barbers DB table.

Page	File	Description
Contact	contact.php	Contact details, Google Maps embed (from settings DB), and contact form.

4. Admin Panel

Admin panel is located at `views/admin/`. Requires login. Session expires after 2 hours of inactivity.

Admin URL	<code>localhost/bg-barbershop/views/admin/login.php</code>
Session	2-hour auto-timeout for security
Access	Admin login required for all pages — redirects to login if not authenticated

4.1 Dashboard (`dashboard.php`)

- Today's bookings count, pending count, completed this month, total clients
- Today's Schedule table — confirm, mark done, or cancel appointments inline
- Recent Bookings table — last 10 bookings with quick status view

4.2 Appointments (`appointments.php`)

- Filter by status (pending/confirmed/completed/cancelled/no_show), date, and search by name/phone/ref
- Actions: Confirm, Mark as Done, Cancel, No Show, Add Notes
- Print button — opens print-ready report in new tab
- Export CSV — downloads spreadsheet with all filtered appointments

4.3 Services (`services.php`)

- View all services grouped by category
- Add new service with category, name, description, price, duration
- Edit or delete existing services
- Toggle active/inactive (inactive services are hidden from booking page)
- Toggle featured flag

4.4 Barbers (`barbers.php`)

- View barber cards with photo, role, and status
- Add/edit/delete barbers with photo URL, social links, bio, and sort order
- Activate/deactivate barbers (inactive barbers are hidden from booking and site)

4.5 Gallery (`gallery.php`)

- View all gallery photos with category filter

- Add photo via URL or file upload
- Toggle featured (featured photos appear on homepage gallery)
- Delete photos

4.6 Reports (reports.php)

- Date range filters: Today, This Week, This Month, Last Month, This Year, Custom
- Summary stats: total bookings, completed, revenue, average per booking
- Charts: Daily bookings line chart, booking status doughnut, busiest days bar chart
- Top services: bar progress with booking count and revenue
- Barber performance table: completion rate, revenue per barber

4.7 Block Dates (blocked_dates.php)

- Block entire shop or specific barber on a date
- Full day block or specific time range (e.g., lunch break)
- Quick presets for Philippine holidays (Christmas, New Year, Independence Day, etc.)
- Blocked dates automatically gray out in booking page time slots

4.8 Walk-in Queue (queue.php)

- Add walk-in customers with name, phone, barber, and service
- Queue status: Waiting → In Progress (In The Chair) → Done
- Shows estimated start time based on queue position
- Shows time in chair for in-progress customers
- Auto-refreshes every 30 seconds
- Clear Done button to remove completed customers

4.9 Reviews (reviews.php)

- Add, edit, delete customer reviews/testimonials
- Star rating input (1–5 stars)
- Toggle featured (featured reviews appear on homepage)
- Toggle active/inactive (hidden reviews not shown on site)

4.10 Customers (customers.php)

- View all registered customers with booking stats
- Search by name, email, or phone
- Filter by active/inactive status

- View customer detail modal with last 5 bookings
- Toggle active/inactive, delete customer

4.11 Settings (settings.php)

Section	Settings Available
General	Shop name, tagline, address, phone, email, weekday/weekend hours
Social Media	Facebook URL, Instagram URL, TikTok URL
Booking	Time slot interval (15/30/45/60 mins), max advance days, Google Maps embed URL
Change Password	Current password verification + new password with strength indicator
System Info	PHP version, MySQL version, server info, logged-in user

5. Email Notification System

Uses PHPMailer 7.0 with Gmail SMTP. Configuration is in config/mail.php.

5.1 Email Configuration

SMTP Host	smtp.gmail.com
Port	587 (STARTTLS)
Auth	Gmail App Password (16 characters, no spaces)
Config File	config/mail.php
Mailer File	config/mailer.php
Min Score	0.5 (reCAPTCHA v3)

Gmail App Password is required. 2-Step Verification must be enabled on the Gmail account. Generate at: myaccount.google.com > Security > App Passwords.

5.2 Email Types

Type	Trigger	Recipients	Function
Booking Confirmation	Customer submits booking form	Customer	sendBookingConfirmation()
Admin Notification	New booking received	Admin email (MAIL_ADMIN)	sendAdminNewBooking()
Cancellation Notice	Booking cancelled by admin or customer	Customer	sendCancellationNotice()
Booking Reminder	Daily cron job at 8:00 AM	Customers with appointment tomorrow	sendBookingReminder()

5.3 Booking Reminder Cron Job

The reminder system uses Windows Task Scheduler to run automatically every morning.

File	Location	Purpose
send_reminders.php	cron/	PHP script that queries appointments tomorrow and sends reminder emails
run_reminders.bat	cron/	Windows batch file that runs PHP via XAMPP

File	Location	Purpose
reminder_log.txt	cron/	Auto-generated log file of all reminder sends

Task Scheduler Setup: Windows Search > Task Scheduler > Create Basic Task > Daily, 8:00 AM > Start Program: C:\xampp\htdocs\bg-barbershop\cron\run_reminders.bat

6. Security Implementation

Security is centralized in `config/security.php`. It is loaded in all login pages and API endpoints.

6.1 Security Features

Feature	Implementation	Protection Against
CSRF Tokens	Random 32-byte token in session, validated on every POST	Cross-site request forgery attacks
Honeypot Fields	Hidden form field — filled by bots, empty for humans	Automated bot submissions
Brute Force Protection	5 failed attempts = 15-minute lockout per email	Password guessing attacks
Rate Limiting	Max 5 bookings per IP/hour, max 10 logins per IP/hour	Spam and flood attacks
Booking Spam	Max 3 bookings per phone number per day	Fake bulk bookings
Input Sanitization	<code>clean()</code> , <code>clean_email()</code> , <code>validate_phone()</code> , <code>validate_date()</code>	SQL injection and XSS attacks
Session Security	Session ID regenerated every 30 mins, admin timeout 2 hours	Session hijacking and fixation
Security Headers	X-Content-Type-Options, X-Frame-Options, X-XSS-Protection	Clickjacking, MIME sniffing, XSS
reCAPTCHA v3	Google reCAPTCHA v3 on booking and login forms — min score 0.5	Bot submissions
Date Validation	Cannot book appointments in the past	Invalid booking attempts
Password Hashing	bcrypt with cost factor 10 (<code>password_hash</code>)	Password database breaches

6.2 reCAPTCHA Configuration

Type	reCAPTCHA v3 (invisible — no user interaction required)
Site Key	6Lef9plsAAAADlpJ9LESMH2t4rZE_qiuDK6P9IT
Domain	localhost (add production domain when deploying)
Min Score	0.5 (0.0 = bot, 1.0 = human)

Pages	Booking form, Customer login, Admin login
Dashboard	google.com/recaptcha/admin

When deploying to a live server, add the production domain in the reCAPTCHA admin dashboard.

7. API Endpoints

7.1 Public APIs

Endpoint	Method	Description
api/book_appointment.php	POST	Submit a new booking. Validates conflicts, checks rate limits, saves to DB, sends confirmation email.
api/get_slots.php	GET	Returns booked slots and blocked ranges for a barber on a given date. Params: barber_id, date.

7.2 Admin APIs

Endpoint	Method	Description
api/admin/update_appointment.php	POST	Update appointment status or admin notes. Sends cancellation email if status = cancelled.
api/admin/export_appointments.php	GET	Export appointments as CSV or print-ready HTML. Params: type, status, from, to.
api/admin/get_customer_bookings.php	GET	Get last 5 bookings for a customer. Param: customer_id.

8. Deployment Guide

8.1 Local (XAMPP) Setup

Step	Action
1	Install XAMPP — ensure Apache and MySQL are running
2	Copy bg-barbershop folder to C:/xampp/htdocs/
3	Open phpMyAdmin at localhost/phpmyadmin
4	Create database: biglang_gwapo
5	Import: biglang-gwapo-db.sql via phpMyAdmin > Import tab
6	Configure: edit config/mail.php with Gmail credentials
7	Install PHPMailer: run "composer require phpmailer/phpmailer" in project folder
8	Access: localhost/bg-barbershop/
9	Admin login: localhost/bg-barbershop/views/admin/login.php

8.2 Live Server Deployment

Step	Action
1	Purchase domain and hosting with PHP 8+ and MySQL support
2	Upload all files via FTP/cPanel File Manager
3	Create MySQL database and import biglang-gwapo-db.sql
4	Update config/db.php with live database credentials
5	Update config/mail.php with production email settings
6	Add production domain to reCAPTCHA admin dashboard
7	Enable HTTPS/SSL (free via Let's Encrypt)
8	Update all hardcoded /bg-barbershop/ paths to production domain

Step	Action
9	Set up cron job on server (replace Windows Task Scheduler)
10	Test all features: booking, email, admin panel

For live deployment, replace all occurrences of "/bg-barbershop/" with the root URL of the production server (e.g., "/") in all PHP files.

8.3 Cron Job for Live Server

Replace Windows Task Scheduler with a server cron job. Add via cPanel or server SSH:

```
0 8 * * * php /path/to/bg-barbershop/cron/send_reminders.php
```

9. Maintenance & Troubleshooting

9.1 Common Issues

Issue	Possible Cause	Solution
Cannot login to admin	Wrong password or locked out	Run reset_admin.php helper file (see documentation note)
Emails not sending	Wrong Gmail credentials or App Password	Verify config/mail.php credentials. Check Gmail 2-Step Verification is ON.
Booking says "Security check failed"	reCAPTCHA token not loading	Check internet connection (reCAPTCHA requires Google CDN)
Gallery images not showing	Broken Unsplash URL	Replace URL in Admin > Gallery or use local image files
Walk-in queue not working	walk_in_queue table missing	Run create_queue_table.php once, then delete file
Reminder emails not sending	XAMPP not running when Task Scheduler fires	Ensure XAMPP is always running or use live server cron
CSS not loading	BASE_PATH incorrect	Check that pages.css and style.css are linked in header.php

9.2 Regular Maintenance Tasks

- Monthly: Review and clear old walk-in queue records from phpMyAdmin
- Monthly: Check cron log file at cron/reminder_log.txt for failures
- Quarterly: Update barber photos and bios in Admin > Barbers
- Quarterly: Review and update service prices in Admin > Services
- Annually: Rotate admin password via Admin > Settings > Change Password
- Before holidays: Add blocked dates in Admin > Block Dates using Philippine holiday presets

9.3 Backup Recommendations

- Database: Export biglang_gwapo database monthly via phpMyAdmin > Export
- Files: Backup assets/images/gallery/ folder if using local image uploads
- Settings: Keep a copy of config/mail.php and config/db.php in a secure location

10. Quick Reference

10.1 Important URLs

Page	URL
Homepage	localhost/bg-barbershop/
Booking Page	localhost/bg-barbershop/views/user/booking.php
Customer Login	localhost/bg-barbershop/views/user/login.php
Customer Register	localhost/bg-barbershop/views/user/register.php
My Bookings	localhost/bg-barbershop/views/user/my_bookings.php
Admin Login	localhost/bg-barbershop/views/admin/login.php
Admin Dashboard	localhost/bg-barbershop/views/admin/dashboard.php
Admin Appointments	localhost/bg-barbershop/views/admin/appointments.php
Admin Reports	localhost/bg-barbershop/views/admin/reports.php
Admin Walk-in Queue	localhost/bg-barbershop/views/admin/queue.php
Admin Settings	localhost/bg-barbershop/views/admin/settings.php

10.2 Key Configuration Files

File	Purpose	What to Edit
config/db.php	Database connection	hostname, username, password, database name
config/mail.php	Email settings	Gmail address, App Password, admin email
config/security.php	Security settings	reCAPTCHA keys, rate limits, lockout duration

10.3 Appointment Status Flow

Status	Description	Next Possible Status
pending	Newly submitted booking	confirmed, cancelled, no_show
confirmed	Admin has confirmed the slot	completed, cancelled, no_show
completed	Service has been rendered	Final — no further changes

Status	Description	Next Possible Status
cancelled	Cancelled by admin or customer	Final — cancellation email sent
no_show	Customer did not arrive	Final — no action

Document End

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For technical support and updates, refer to the developer who built this system.